

Equity Valuation



Divija Balasankula

Ray Cui

Jinjia Peng

Isaac Gohn

MBA 524

October 28, 2025

Table of Contents

I.	Industry Analysis.....	p. 3-6
a.	Overview	
b.	Porter's 5 Competitive Forces	
c.	Expected Future Trends	
II.	Stock Screen & Alternative Stock Selection.....	p. 7
III.	Snapshot Descriptions.....	p. 7-10
a.	General Motors (GM)	
b.	Honda Motor Co. (HMC)	
c.	Toyota Motor Corp. (TM)	
IV.	Alternative Stock Selection – Final Selection.....	p. 11
V.	Company Specific Reports – Tesla.....	p. 12-20
a.	Investment Thesis	
b.	Overview & Discussion of Valuation	
c.	Sensitivity Analysis	
d.	Scenario Analysis	
e.	Sensitivity/Scenario Analysis Conclusion	
VI.	Company Specific Reports – General Motors.....	p. 21-28
a.	Investment Thesis	
b.	Overview & Discussion of Valuation	
c.	Sensitivity Analysis	
d.	Scenario Analysis	
e.	Sensitivity/Scenario Analysis Conclusion	
VII.	Summary & Recommendation.....	p. 28-29
VIII.	References.....	p. 30

I. Industry Analysis

Overview

The automobile manufacturing industry is one of the largest sectors in the global economy, producing more than 90 million vehicles annually and generating revenues of over \$3.5 trillion. It encompasses the design, engineering, production, and distribution of passenger cars, commercial vehicles, and increasingly, electric vehicles (EVs). The industry also depends on a vast network of suppliers providing raw materials, parts, electronics, and software, making it highly integrated and complex.

Historically dominated by giants such as Toyota, Volkswagen, General Motors, Ford, Honda, and Hyundai-Kia, the competitive landscape is shifting as new players, particularly in the EV space, rise in importance. Companies like Tesla, BYD, and SAIC are reshaping the industry through electrification, digitalization, and direct-to-consumer business models. Traditional automakers are investing billions in battery technology, autonomous driving systems, and connected car platforms to remain competitive in this evolving environment.

Some defining characteristic of the industry is its globalized structure. Production is spread across North America, Europe, and Asia, with China now the largest car market and production hub. Trade policies, tariffs, and geopolitical tensions have significant effects on manufacturers' costs and strategies. In recent years, companies have also moved toward regionalized supply chains to mitigate risks from global disruptions such as the COVID-19 pandemic and the semiconductor shortage.

Technology is rapidly transforming automobiles from purely mechanical products into "software-defined vehicles." Features like over-the-air updates, advanced driver-assistance systems (ADAS), and vehicle connectivity are becoming standard, creating new revenue streams for manufacturers. Partnerships with technology firms, battery producers, and chipmakers are increasingly vital. Furthermore, autonomous driving research, though still in early stages, holds the potential to radically alter how vehicles are used and owned.

Future trends point toward a transformation from hardware-focused manufacturers to "mobility companies" that generate revenue not just from vehicle sales but also from software, subscription services, and shared mobility platforms. By 2035, EVs are expected to represent nearly half of all new car sales globally, with autonomous and connected vehicle technologies further redefining the sector. Sustainability commitments are pushing manufacturers toward carbon neutrality in both production and product lifecycles, adding pressure but also creating opportunities for innovation.

In sum, the automobile manufacturing industry stands at a pivotal moment: balancing traditional mass production with the urgent push toward electrification, sustainability, and technological innovation. Its long-term success will depend on how effectively companies adapt to these changes while navigating the risks of a highly competitive and rapidly evolving global marketplace.

Porter's 5 Forces

Porter's Five Forces is a framework used to understand how competition shapes an industry and where profits may come from. It examines five pressures: the intensity of rivalry among existing competitors, the threat of new entrants, the threat of substitutes, the bargaining power of suppliers, and the bargaining power of buyers. Looking at each of these forces makes it easier to see the pressures on profitability and the opportunities that might exist within the market.

High Intensity of Rivalry Among Existing Competitors

The automotive industry is highly technical and competitive. First, companies in this industry are investing their capital heavily in areas such as advanced batteries, autonomous driving, etc.. Each of the competitors strives to provide the best new technologies for their customers, creating more competition in the market. At the same time, emerging technologies such as solid-state batteries, autonomous driving, and connected vehicle platforms are shifting the competitive landscape and forcing incumbents to innovate at a faster pace. On a global scale, the competition is further intensified because of the new entrants. For instance, China's leading EV firms BYD, NIO, and XPeng are bringing new innovations to the market. Their cost-efficient production and unique approaches such as battery-swapping infrastructure further stimulates the market's competitiveness.

Increasing Threat of New Entrants

The threat of new entrants is moderate but rising as market conditions change. In the past, factors such as large capital requirements, complex supply chains, and regulatory policies served as barriers that blocked the entry of new firms. The challenges still remain today; however, factors such as falling battery costs, modular assembly lines, and contract manufacturing are causing lowering such barriers. As mentioned earlier, the global rise of Chinese EV makers illustrates how new firms supported by scale and innovation are able to compete effectively. This phenomenon suggests that the threat of new entrants is increasing and will continue to pressure established automakers over a longer period.

Moderate Threat of Substitutes

The threat of substitutes in the automobile industry is moderate. It is shaped by both alternative technologies and mobility trends. Traditional substitutes include public transit, ride-hailing, and micromobility options such as bikes and scooters, still remain in the market. However, the emerging new automobile technology innovation such as the hydrogen fuel cells and hybrid

vehicles offer consumers alternative paths to low-emission transport alongside full electric vehicles. While these alternatives have not yet been fully implemented inside the market, their influence is likely to grow in the future as they become more mature. Therefore, the level of threat of substitutes is considered moderate.

High Bargaining Power of Suppliers

Looking at the market, batteries, semiconductors, and raw materials such as lithium, cobalt, and nickel are critical, and the markets for these inputs are highly concentrated. In addition, some raw materials are only provided in certain countries, making it difficult to obtain if relative policies change. The core supplier's effect is also significant. For example, battery producers, in particular, dominate global supply and influence prices and production allocation. Shortage in battery production would pause thousands of cars' manufacturing process. As mentioned early, the intense rivalry among automakers only increases the pressure to secure reliable access to these critical inputs. Although companies are pursuing vertical integration and supply diversification, supplier leverage remains structurally high in the near term. Having said that, the bargaining power of suppliers remains high for now and the future.

Moderate Bargaining Power of Buyers

Buyers exert moderate bargaining power in the automobile industry. In general, consumers have multiple choices such as gasoline, hybrid, and electric based models. This creates a strong bargaining power and significantly lowers the cost of exit. It is a trend that the future automobile model choices vary, as new usable energy is discovered or proven reliable. However, companies also have created new strategies such as common digital platforms and over-the-air system updates to raise switching costs. Considering these facts, the bargaining power of buyers remains moderate.

Expected Future Trends

One future trend is the discovery of new energy sources. This trend would significantly affect these five forces. First, between each competitor, firms will compete for proprietary technologies that could utilize the new resources. This would increase the rivalry among existing competitors. It will also increase the threat of the new entrants, since they might have core technologies to take away the old positions of the existing firms. The threat of substitutes will be lowered because a leading new energy source would shift the market focus, and thus reducing profitability of the old resources. The threat of suppliers would be higher initially because the new energy might be scarce to the market, and it might get shifted down as time moves forward. Finally, if the new emerging energy source becomes the major part of the market, the bargaining power of the buyers will be lowered because the new energy source dominates the market and there isn't much of replacement.

To summarize the trend, the automotive industry will continue to see rising competition. These technological races, combined with the entry of aggressive global competitors such as the Chinese EV makers, will compress margins and force automakers to constantly innovate to protect their market share. This competition should be treated as good circulation in the market, which drives this industry forward with new innovations and market expectations. In the future, as new energy sources become available, traditional automakers will be forced to diversify their products, and the competition will become more intensified.

II. Stock Screen and Alternative Stock Selection

To identify potential alternatives for SunTrust portfolio holding, our team conducted a structured stock screening based on YChart. To guarantee liquidity, the first step was to ensure all candidates traded on major U.S. exchanges and had a market capitalization of at least one billion dollars. Based on this board set, we selected candidates strictly within the Automobiles industry from major GICS sectors. Then, we utilized additional valuation and quality factors to refine the list further. Specifically, we examined valuation multiples such as PE, PS, and EV/EBITDA and also placed emphasis on profitability factors such as operating margins and gross margin. We also reviewed free cash flow to emphasize the capability of firms internal funding growth.

Through this process, four companies emerged as the most compelling alternatives: **Ferrari (RACE)**, **General Motors (GM)**, **Honda Motor Co. (HMC)**, and **Thor Industries (THO)**. Ferrari (RACE) was selected in our alternatives because of its industry-leading gross and relatively strong operating margins. Its operating margin is about 31% and its gross margin is above 50% which are far higher than most car makers. These numbers show Ferrari can charge premium prices and keep costs under control, although its P/E of around 49 makes it more expensive than peers. General Motors was included because it offers strong value. GM trades at a P/E of only 9 and a P/S ratio close to 0.3, much lower than the industry average (P/E 10 - 15). Its operating margin is modest at about 4.5%, but it is positive, which makes the low valuation attractive. Honda Motor Co. provides a similar value story. Its P/E is about 11 and P/S around 0.35. It also shows a stronger gross margin of 20%, which is higher than GM, giving it more stability.

Symbol	Company	Exchange	Industry	Market Cap.(\$B)	P/E	P/S	EV/EBITDA	Gross Margin (%)	Operating Margin (%)	ROE (%)	ROA (%)	Revenue YoY (%)	FCF (\$M)
TSLA	Tesla, Inc.	NASDAQ	Automobiles	1277.1	228.9	15	109	17.2	4.1	8.4	5	-11.8	5586
TM	Toyota Motor Corp.	New York Stock Exchange	Automobiles	255.6	9.1	0.79	10.2	17.9	9.5	11.8	4.6	11.6	-603
RACE	Ferrari NV	New York Stock Exchange	Automobiles	85.2	49.3	11.31	29.6	52.6	31.1	46.4	16.9	10	1491
GM	General Motors Co.	New York Stock Exchange	Automobiles	55.7	9.3	0.32	5.5	9.1	4.5	7	1.7	-1.8	-1843
HMC	Honda Motor Co., Ltd.	New York Stock Exchange	Automobiles	44.6	11.6	0.35	5.8	20.1	4.6	5.5	2.3	6.6	-2568

III. Snapshot Descriptions

General Motors (GM)

General Motors Company, founded in 1908 by William C. Durant in Detroit, Michigan, is one of America’s largest automakers and has been the top-selling automaker in the United States for decades. The company operates through four main segments: GM North America, GM International, Cruise (autonomous vehicles), and GM Financial. GM’s portfolio includes four iconic brands: Chevrolet, Buick, GMC, and Cadillac, each targeting different market segments from mainstream to luxury vehicles.

As of 2024, GM ranks 25th by total revenue among American companies on the Fortune 500 and 50th globally. The company has announced ambitious plans to achieve carbon neutrality by 2040 and end production of internal combustion engine vehicles by 2035, though these plans were scaled back in 2025. GM operates manufacturing facilities across eight countries and employs over 90,000 people globally.

GM News

In January 2025, GM reported strong Q4 2024 earnings that exceeded analyst expectations with adjusted EPS of \$1.92 versus the estimated \$1.89, and revenue of \$47.7 billion representing an 11% increase year-over-year. The company also provided optimistic 2025 guidance with projected net income of \$11-12 billion and adjusted EBIT of \$13-15 billion, despite investor concerns about potential Trump administration policy changes. For the full year 2024, GM achieved record performance with adjusted EBIT of \$14.9 billion and adjusted automotive free cash flow of \$14 billion, demonstrating strong growth in both electric vehicle and traditional ICE sectors.

Financial Metrics (as of September 2025)

GM

Previous Close	58.71	Market Cap (intraday)	55.697B
Open	59.85	Beta (5Y Monthly)	1.32
Bid	58.26 x 600	PE Ratio (TTM)	8.93
Ask	58.60 x 400	EPS (TTM)	6.55
Day's Range	58.47 - 60.00	Earnings Date	Oct 21, 2025
52 Week Range	41.60 - 61.24	Forward Dividend & Yield	0.60 (1.03%)
Volume	7,819,768	Ex-Dividend Date	Sep 5, 2025
Avg. Volume	8,690,583	1y Target Est	58.28



Honda Motor Co., Ltd. (HMC)

Honda Motor Co., Ltd., founded in 1946 by Soichiro Honda in Tokyo, Japan, is a leading global manufacturer operating through four primary business segments: Motorcycle Business, Automobile Business, Financial Services Business, and Power Product and Other Businesses. The company has grown from its origins as a motorcycle manufacturer to become one of the world's largest automotive companies, known for reliability, fuel efficiency, and innovation.

Honda's automotive portfolio includes passenger cars, light trucks, and mini vehicles under both the Honda and luxury Lexus brands. The company is particularly recognized for popular models like the Civic, Accord, CR-V, and Pilot. Beyond automobiles, Honda maintains its position as the world's largest motorcycle manufacturer and produces a diverse range of power products including generators, lawn mowers, and marine engines. The company also manufactures the HondaJet aircraft, demonstrating its technological versatility.

Honda (HMC) News

In August 2025, Honda reported strong Q1 FY2026 results with operating profit of 244.1 billion yen, driven by record motorcycle business performance in Brazil and Vietnam, while maintaining strong North American auto sales. The company also revised its FY2026 forecast upward to operating profit of 700 billion yen and net profit of 420 billion yen, representing a 200 billion yen increase from previous guidance. Earlier in February 2025, Honda announced significant progress on its 1.1 trillion yen share buyback program, having acquired 936.5 billion yen (85.1% completion) by July 2025.

Financial Metrics (as of September 2025)

HMC

Previous Close	33.96	Market Cap (intraday)	43.868B
Open	33.63	Beta (5Y Monthly)	0.32
Bid	33.02 x 1300	PE Ratio (TTM)	11.44
Ask	33.86 x 200	EPS (TTM)	2.93
Day's Range	33.50 - 33.70	Earnings Date	Aug 6, 2025
52 Week Range	23.41 - 34.89	Forward Dividend & Yield	1.42 (4.24%)
Volume	421,835	Ex-Dividend Date	Mar 28, 2025
Avg. Volume	964,780	1y Target Est	37.19



Toyota Motor Corporation (TM)

Toyota Motor Corporation, established in 1937 and headquartered in Japan, is one of the world's largest automotive manufacturers by sales volume and revenue. The company operates through three main business segments: Automotive (91.8% of revenue), Financial Services (8.2% of revenue), and All Other businesses. Toyota's automotive portfolio spans the full spectrum from subcompact cars to luxury vehicles, commercial trucks, and buses under the Toyota, Lexus, Daihatsu, and Hino brands.

Toyota is renowned for pioneering hybrid technology with the Prius, the world's first mass-produced hybrid vehicle, and continues to lead in fuel-efficient vehicle technology. The company sells vehicles in approximately 190 countries and regions, with North America being its largest market outside Japan. Toyota has committed to offering electrified models across its entire lineup by 2025, reflecting its strategic focus on environmental sustainability.

Toyota (TM) News

In 2024, Toyota reported record revenue of approximately \$256 billion (¥37 trillion) for FY2023, marking a 15.4% increase from the previous year. The company sold 8.8 million vehicles globally, with 76% of sales occurring outside Japan, demonstrating its strong international presence. Toyota continues to invest heavily in R&D and maintains research centers across the United States, Japan, China, and Europe focused on advanced vehicle technologies.

Financial Metrics (as of September 2025)

TM

Previous Close	197.63	Market Cap (intraday)	255.752B
Open	195.76	Beta (5Y Monthly)	0.22
Bid	195.33 x 100	PE Ratio (TTM)	8.89
Ask	196.50 x 100	EPS (TTM)	22.06
Day's Range	195.42 - 196.35	Earnings Date	Aug 7, 2025
52 Week Range	155.00 - 202.85	Forward Dividend & Yield	6.91 (3.53%)
Volume	163,247	Ex-Dividend Date	Mar 31, 2025
Avg. Volume	416,125	1y Target Est	235.27



IV. Alternative Stock Selection

Tesla occupies a unique position as both a technology company and an automotive manufacturer, so potential portfolio replacements extend beyond traditional automakers to include both established OEMs transitioning to electrification and pure-play EV specialists that excel in autonomous driving, energy storage, and sustainable transportation. Among legacy OEMs, General Motors has pledged \$35 billion to EV development through 2025 and aims for an all-electric lineup by 2035; Ford's Lightning electric truck and Mustang Mach-E have driven strong U.S. demand and demonstrated the company's ability to leverage existing manufacturing scale; and Toyota plans 15 new EV models with the capacity to produce 1.5 million electric vehicles annually by 2026, signaling its shift from hybrid leadership to full electrification. Emerging global EV competitors from China - BYD, NIO, Li Auto, and XPeng - have captured significant market share in the world's largest EV market through innovative, cost-efficient production and integrated supply chains; BYD alone generated over \$107 billion in 2024 revenues, surpassing Tesla's \$97.7 billion and benefiting from its vertically integrated battery and semiconductor operations. U.S. startups Rivian and Lucid, targeting premium truck and luxury sedan segments respectively, offer compelling growth prospects but face challenges scaling production and achieving sustained profitability. In the luxury automotive segment, Ferrari stands out with plans to launch its first all-electric supercar in 2025 and has maintained industry-leading profitability margins above 28%, providing differentiated exposure to the electrification theme without competing in the mass-market EV arena.

Final Selection: General Motors

After eliminating firms with insufficient scale, weak financial stability, or high regulatory risks, General Motors emerged as the primary alternative. GM currently stands as the second-largest EV manufacturer in the United States, offering 13 EV models alongside established production and distribution infrastructure. The company combines cash flow generation and dividend payments with attractive valuation multiples compared to Tesla. GM's Q4 2024 results demonstrated resilience, with revenue up 11% year-over-year to \$47.7 billion and net income guidance for 2025 projected between \$11-12 billion. Unlike unprofitable EV startups, GM provides exposure to the EV transition while maintaining the stability of a mature automotive manufacturer. This balance of growth potential and reduced volatility makes GM a suitable replacement for Tesla within the portfolio.

V. Company Specific Reports - Tesla

Investment Thesis

Tesla Inc. is the world's leading manufacturer of electric vehicles. The main models it produces include Model 3, Model Y, Model S, Model X, and Cybertruck, and it also offers energy storage systems and solar products. Despite maintaining a technological leadership and innovation advantage in the market, our valuation analysis indicates that Tesla's stock price is significantly overvalued. According to the equity free cash flow (FCFE) model, the intrinsic value per share of Tesla is only \$12.94, which is approximately 97% lower than the current market price of \$403.84. The company's current price-to-earnings ratio (P/E) is 233 times, and the price-to-book ratio (P/B) is 16.4 times. These extremely high valuation levels cannot be reasonably explained by its expected cash flows.

Our model predicts that Tesla's average annual sales growth rate over the next five years will be approximately 6.5%, and the cost of equity will be 18.4%. Based on these assumptions, the FCFE model calculates the intrinsic value per share to be \$12.94, and the current price indicates that the stock trading price is approximately 31 times its intrinsic value. The company's cost structure shows a high operational burden. The cost of goods sold (COGS) accounts for 76.6% of sales revenue, and selling and administrative expenses (SG&A) account for 9.9%, both higher than those of traditional automotive manufacturers. It is expected that the company's earnings before interest and taxes (EBIT) will increase from \$7.8 billion to \$9.3 billion over the next five years. Such growth is insufficient to support the current market capitalization of \$1.3 trillion.

Globally, Tesla's market share has dropped from the initial 70% to approximately 39%. The main reason for this decline is the sharp intensification of competition. In the Chinese market, BYD has surpassed Tesla in terms of electric vehicle sales. In the US market, General Motors has launched 13 electric vehicle models, while traditional automakers such as Ford and Volkswagen are also accelerating the launch of their own electric vehicle products. Intense competition has forced Tesla to repeatedly lower prices to stimulate demand, which has temporarily boosted sales but significantly reduced profit margins, reflecting the increasingly commoditized trend in the electric vehicle market.

The capital-intensive nature of Tesla further reduces the expansion space of its free cash flow. The ratio of the company's fixed assets to sales is as high as 72%, indicating that its business is highly dependent on continuous large-scale investment. Compared to a market capitalization of 1.3 trillion US dollars, the company's free cash flow is still limited. Tesla does not pay any dividends, with a dividend payout ratio of 0%, and shareholders rely entirely on stock price increases to obtain returns. It is expected that the cumulative depreciation of the company will increase from 189 billion US dollars to 466 billion US dollars within the next five years. This trend reflects that the company still needs to make continuous capital investments to maintain its production capacity and technological leadership.

The scenario analysis results were all unsatisfactory under all assumptions. The valuation under the base scenario was \$12.94 per share, with a 97% downside potential; in the pessimistic scenario, assuming a 3% sales growth and an 23% equity cost, the valuation was \$3.61, with a decline of up to 99%; in the optimistic scenario, assuming a 9% sales growth and an 15% equity cost, the valuation was \$16.51, still lower than the current market price by 96%. Even in the most optimistic scenario, Tesla's stock price is still grossly overvalued, indicating that its current valuation has far exceeded the reasonable range.

Apart from the valuation issue, the company also faces multiple risks at the operational level. The production progress of Cybertruck is constrained by rising costs and capacity bottlenecks, and the fully autonomous driving (FSD) function has not yet received regulatory approval. The CEO Elon Musk is simultaneously managing multiple enterprises (including SpaceX, X, Neuralink, etc.), which has raised questions about his focus on Tesla's core business. At the same time, product quality issues and recall incidents occur frequently, potentially affecting the brand reputation. As the electric vehicle industry gradually enters a mature stage, with converging technologies, an increase in competitors, and weakened cost advantages, Tesla's growth potential is being eroded.

Nevertheless, we acknowledge that Tesla still has some potential for upward growth, such as significant breakthroughs in autonomous driving technology or rapid expansion of its energy storage business. However, these scenarios are highly uncertain and are constrained by external factors such as technical difficulty and

regulatory environment. The likelihood of their realization is extremely low. The current market valuation already incorporates these idealized growth assumptions, lacking sufficient real-world support.

In conclusion, we recommend selling Tesla stocks. There is a significant disconnect between Tesla's current stock price and its fundamentals. It has 97% of its downside potential, a P/E ratio of 233, a market share dropping from 70% to 39%, a sales growth rate insufficient to support the valuation, no dividends paid, and negative results in all scenario analyses, all indicating that the current risk-reward ratio of the company is extremely unreasonable. Tesla's stock price is mainly driven by market sentiment and investment speculation rather than by real cash flow and corporate profitability. At the current price level, the investment risk of Tesla is significantly higher than the potential return.

Overview and Discussion of Valuation

Cost of capital

We used the Fama–French Three-Factor Model to estimate the cost of equity. We used it over CAPM because it only considers market risk, but real-world returns are also affected by company size and value characteristics. The Fama–French model adds these dimensions to better capture systematic risk. The formula is as follows:

$$r_i = R_F + \beta_i^{\text{mkt}} \text{RMRF} + \beta_i^{\text{size}} \text{SMB} + \beta_i^{\text{value}} \text{HML},$$

Where:

- R_F is the Risk Free Rate of return.
- RMRF is the excess market return.
- SMB (Small Minus Big) is the return difference between small and large firms.
- HML (High Minus Low) is the return difference between high and low book-to-market firms (value effect).
- The betas are the sensitivities for each of the factors.

Tesla		
factors	Annual factors	Beta sensitivities
RMRF	.1035	1.407
SMB	.027	1.325
HML	.0509	-1.168
Risk Free Rate	.0332	

To calculate these terms, we used data from 1926 to 2024 and the one-month U.S. Treasury bill rate as the risk-free rate.

The high market beta shows that Tesla's returns are 41% more volatile than the overall market, consistent with its aggressive growth and sensitivity to investor sentiment. The positive SMB beta indicates Tesla behaves like a smaller, fast-growing company despite its large capitalization, while the negative HML beta aligns with Tesla being a growth stock rather than a value stock. Overall, the 23.65% cost of equity reflects the premium investors demand for Tesla's high risk and uncertain cash flows.

Because Tesla does not pay dividends, we applied the Free Cash Flow to Equity (FCFE) valuation model. This model estimates the cash available to shareholders after accounting for capital expenditures, working capital needs, and debt servicing—an appropriate framework for Tesla given its reinvestment-heavy growth strategy.

Our model used the following assumptions:

Assumptions Block		
Choose these yourself		
Sales growth per year for next 5 years	6.55%	Gross Fixed Assets (PPE)/Sales. 72.06%
Current Stock Price	403.84	Depreciation/Trailing Gross Fixed Assets 8.92%
Long Term Growth Rate in FCFE	5%	Current assets/Sales 59.74%
Long Term Growth Rate in Dividends	0%	Current liabilities/Sales 29.50%
Terminal P/E Value	318	COGS/Sales 76.64%
Buy/Sell/Hold Margin	10%	SGA/Sales 9.92%
Choice of Terminal Value for FCFE Model: 1= Gordon Growth, 2= Terminal PE model	1	Interest /Trailing Debt 5.36%
Choice of Terminal Value for Dividend Model: 1= Gordon Growth, 2= Terminal PE model	2	Dividend payout % of earnings 0.00%
		Tax Rate 24.79%
		Fixed Debt/Assets Ratio 8.74%
		All Other Assets (Assume stays fixed \$ amount) \$8,728.00
		Shares Outstanding 3216 f

Our projections indicate Tesla's sales are expected to grow at approximately 6.5% per year, increasing from \$104 billion in 2025 to \$134 billion in 2029. EBIT rises from \$7.7 billion to nearly \$10 billion, while FCFE grows steadily from \$5.3 billion in 2025 to \$6.9 billion in 2029.

This growth reflects modest improvement in margins and production scaling but also recognizes rising costs related to R&D, competition, and infrastructure expansion. Tesla's 0% dividend payout ratio supports reinvestment in future technologies such as energy storage, robotics, and autonomous driving.

	2025	2026	2027	2028	2029
	1	2	3	4	5
Sales	104086.71	110902.28	118164.12	125901.47	134145.46
COGS	79774.60	84998.21	90563.87	96493.96	102812.35
SGA	10324.50	11000.54	11720.86	12488.33	13306.07
Depreciation	6277.13	6688.15	7126.09	7592.70	8089.87
EBIT	7710.49	8215.37	8753.31	9326.48	9937.17
Interest	555.45	565.54	576.29	587.75	599.95
Earnings before Tax	7155.04	7649.83	8177.02	8738.73	9337.22
Tax	1773.79	1896.46	2027.15	2166.40	2314.77
Earnings after Tax	5381.24	5753.37	6149.87	6572.33	7022.45
Dividends	0.00	0.00	0.00	0.00	0.00
Contribution to Retained earnings	5381.24	5753.37	6149.87	6572.33	7022.45

FCFE Valuation								
		Year Relative to Today	0	1	2	3	4	5
Determinants of FCFE								
EBIT*(1-T)				5799.00	6178.71	6583.29	7014.37	7473.66
Add Dep				6277.13	6688.15	7126.09	7592.70	8089.87
Less change in Gross Fixed Assets				4609.70	4911.55	5233.15	5575.82	5940.92
Less Change in NWC				1934.20	2060.86	2195.80	2339.58	2492.78
FCFF				5532.21	5894.46	6280.43	6691.67	7129.84
Less After tax interest payments				417.75	425.34	433.42	442.04	451.22
Add Net Borrowing				188.17	200.49	213.62	227.61	242.51
FCFE				5302.63	5669.62	6060.63	6477.24	6921.14
PV of Cash Flows				4,476.91	4,041.35	3,647.35	3,291.06	2,969.00

Despite optimistic forecasts and strong market positioning, the model's output of \$12.94 per share implies that Tesla's current valuation is not supported by its fundamental cash flows. The result suggests Tesla is overvalued based on its intrinsic value, leading to a Sell recommendation.

Terminal Value (FCFE infinite growth model)	54,054.82
PV of Terminal Value (FCFE infinite growth model)	23,188.20
Terminal Value (Terminal P/E Model)	2,233,138.36
PV of Terminal Value (Terminal P/E Model)	957,961.79
PV Terminal Value	23,188.20
Sum of PVs (Total Intrinsic Value of Equity)	41,613.86
Shares outstanding	3,216.00
Value per share	\$ 12.94
Buy/Sell/Hold	Sell

Sensitivity Analysis

To further test the valuation results under extreme assumptions such as long-term growth rates, sales growth, dividend growth, etc., a sensitivity analysis is

performed. It provides a range of numbers in extreme cases to test different inputs in the valuation model, and generates different results to indicate the effects.

The first valuation being performed is the analysis on the Long-Term FCFE Growth Rate. Observe that fixing a certain rate, as the cost of equity increases, the fair value under FCFE model decreases since the values are being discounted harder. As the Long-Term FCFE Growth Rate increases, the value increases, as a reflection of the positive relationship between the value and growth. Note that the cell that is being highlighted blue is the original model value before the sensitivity tests, and if the extreme value exceeds the current market price, then these values will be highlighted in green.

The values are between 3.94 and 19.44, reflecting the fair value range for Tesla under FCFE model. The increase in the value is gradual(around 2), as the Long-Term FCFE Growth Rate, reflecting a mild sensitivity.

DATA TABLE: SENSITIVITY OF FCFE STOCK VALUATION TO Cost of Equity AND LONG-TERM GROWTH RATE							
		Long-term FCFE growth ↓					
		3%	5%	6%	7%	8%	9%
	12.94						
	15%	9.18	11.23	12.60	14.31	16.51	19.44
	16%	8.12	9.78	10.86	12.18	13.83	15.96
	17%	7.22	8.59	9.46	10.50	11.78	13.37
Cost of Equity Capital	18%	6.46	7.60	8.31	9.15	10.16	11.39
	19%	5.81	6.76	7.35	8.04	8.85	9.83
	20%	5.24	6.05	6.55	7.12	7.78	8.57
	21%	4.75	5.45	5.86	6.34	6.89	7.54
	22%	4.32	4.92	5.28	5.68	6.14	6.68
	23%	3.94	4.46	4.77	5.11	5.50	5.95

Below is the table of the relationship between Cost of Capital and the Long-term Growth Rate of the dividend inside the Divided Discount Model (DDM). Note that since Tesla does not pay dividends, then all valuations here will have zero in value.

		DATA TABLE: SENSITIVITY OF Dividend STOCK VALUATION TO Cost of Equity AND LONG-TERM GROWTH RATE					
	297.87	0%	3%	4%	5%	6%	7%
	15%	0.00	0.00	0.00	0.00	0.00	0.00
	16%	0.00	0.00	0.00	0.00	0.00	0.00
	17%	0.00	0.00	0.00	0.00	0.00	0.00
Cost of Equity Capital	18%	0.00	0.00	0.00	0.00	0.00	0.00
	19%	0.00	0.00	0.00	0.00	0.00	0.00
	20%	0.00	0.00	0.00	0.00	0.00	0.00
	21%	0.00	0.00	0.00	0.00	0.00	0.00
	22%	0.00	0.00	0.00	0.00	0.00	0.00
	23%	0.00	0.00	0.00	0.00	0.00	0.00

The next graph would be how FCFE's valuation changes with the Gross Fixed Assets (PPE) - Sales ratio. As the ratio increases (increase in PPE or decrease in sales), the value of FCFE decreases.

		FCFE Value
		12.94
	10%	9.48674306
Gross Fixed Assets (PPE)/Sales	15%	8.490541667
	20%	7.494340273
	25%	6.49813888
	30%	5.501937487
	35%	4.505736093
	40%	3.5095347
	45%	2.513333306
	50%	1.517131913
	55%	0.5209305196

Scenario Analysis

To perform the scenario, the focus is on the three key valuation drivers: sales growth, long-term FCFE growth, and cost of equity. To perform this scenario analysis, some important business assumptions are being made. Looking at the table, the best-case scenario assumes strong sales growth of 9 percent, long-term FCFE growth of 8 percent, and a lower cost of equity at 15 percent, in which it reflects Tesla's ability to sustain rapid adoption of electric vehicles and its expansion into energy and software. Conversely, the worst-case scenario assumes sales growth of only 3 percent, long-term FCFE growth of 2 percent, and a higher

cost of equity at 23 percent, in which it reflects the potential impact of slowing demand and intensified competition in addition to higher market risk. The simulation results are shown in the below table:

Scenario Summary				
	Current Values:		Worst Case	Best Case
Changing Cells:				
sales_growth	7.00%	3.00%	9.00%	
LTgrowth_FCFE	3%	2%	8%	
cost_equity	18.44%	23.00%	15.00%	
Result Cells:				
Intrinsic_FCFE	12.94	\$ 3.61	\$ 16.51	
Intrinsic_DDM	\$ -	\$ -	\$ -	

Sensitivity/Scenario Analysis Conclusion

A clear conclusion made from the table results is that Tesla is extremely overpriced because all extreme numbers simulated do not exceed the current market price. In general, the result should suggest an immediate short in its equity. However, Tesla's classification is automobile with high tech, as a result, it is normal to have extremely high P/E ratio. Therefore, the long/short recommendation should be re-evaluated thoroughly with other factors, in which it will be discussed later in the final valuation recommendation. The results alone do not contribute to the confidence of the valuation as the data deviates too much from the actual market price. To further test the model, an analysis with a larger number scale should be conducted to reflect this high P/E ratio.

VI. Company Specific Reports - General Motors

Investment Thesis

Our valuation analysis indicates that General Motors (referred to as GM) is currently undervalued. Through the equity free cash flow (FCFE) model, the intrinsic value per share of the company is calculated to be \$96.87, which is 82% higher than the current stock price of \$53.27. The dividend discount model (DDM) estimates the intrinsic value to be \$65.81, corresponding to a 24% upside potential. The results of both valuation models support the "buy" recommendation.

The current price-to-earnings ratio (P/E) of GM is 4.65 times, significantly lower than the industry average of 10-15 times. This discount still exists even when the company's financial performance is stable. In the fourth quarter of 2024, the company's adjusted earnings per share (EPS) was \$1.92 (higher than the market expectation of \$1.89), with revenue reaching \$47.7 billion, a year-on-year increase of 11%. In 2024 as a whole, the company achieved an adjusted earnings before interest and taxes (EBIT) of \$14.9 billion and free cash flow from the automotive business of \$14 billion, demonstrating strong profitability and cash generation capabilities.

Our valuation assumptions remain conservative. We expect the average annual growth rate of sales over the next five years to be only 0.88%, which is lower than the industry average expectation. The cost of equity calculated based on the Fama-French three-factor model is 20.87%. Even under this conservative setting, the model still shows significant underestimation potential. The company's cost of sales (COGS) accounts for 77.6% of sales revenue, and sales and administrative expenses (SG&A) account for 5.0%, demonstrating efficient cost control. The debt-to-asset ratio is 41.2%, which is at a healthy level within the industry.

GM is the second-largest electric vehicle manufacturer in the United States, having launched 13 electric vehicle models so far. The company plans to invest 35 billion US dollars in the fields of electrification and autonomous driving by 2025. This strategy enables the company to maintain profitability of its traditional fuel vehicles while steadily expanding the production scale of electric vehicles. The

company's dividend payout ratio is 5.7%, providing stable returns to shareholders while also retaining sufficient funds for reinvestment and future growth.

The scenario analysis results show that under the base scenario, the FCFE valuation is \$96.87; under the pessimistic scenario (with a 1% increase in sales and a 25% increase in equity cost), the valuation is \$26.76; and under the optimistic scenario (with a 7% increase in sales and a 17% increase in equity cost), the valuation is \$106.23. In the most unfavorable scenario, the stock price could drop by approximately 50%, while in the best scenario, there is a 99% upside potential. This asymmetric risk-reward structure provides strong support for the "buy" recommendation.

The main risks include: the execution risk of the electric vehicle transformation, the competitive pressure from Chinese manufacturers, the impact of government policy changes on the incentives for electric vehicles, and the sensitivity brought about by the fluctuations in the macroeconomic cycle. The currently high equity cost reflects the above market concerns, but as GM gradually proves the profitability of its electric vehicle business, this cost is expected to gradually decrease.

In conclusion, we recommend purchasing General Motors' stocks (GM). The main reasons are as follows: there is an 82% upside potential compared to the intrinsic value; the price-to-earnings ratio is only 4.65 times, significantly lower than the industry average; conservative growth assumptions provide downside protection for the valuation; the 5.7% dividend yield offers stable returns for investors; and the strong ability to generate free cash flow.

Overview and Discussion of Valuation

Cost of capital

To determine GM's cost of equity, we applied the Fama French Three-Factor Model (FFM) the same framework used in the Tesla valuation, to maintain consistency across our analysis.

GM		
factors	Annual factors	Beta sensitivities
RMRF	.1035	1.311
SMB	.027	.563
HML	.0509	.592
Risk Free Rate	.0332	

Assumptions Block			
		Gross Fixed Assets (PPE)/Sales.	68.20%
Choose these yourself		Depreciation/Trailing Gross Fixed Assets	9.15%
Sales growth per year for next 5 years	0.88%	Current assets/Sales	57.91%
Current Stock Price	53.27	Current liabilities/Sales	51.36%
Long Term Growth Rate in FCFE	3%	COGS/Sales	77.62%
Long Term Growth Rate in Dividend	9%	SGA/Sales	5.03%
Terminal P/E Value	13		
Buy/Sell/Hold Margin	10%	Interest /Trailing Debt	8.47%
Choice of Terminal Value for FCFE			
Model: 1= Gordon Growth, 2=		Dividend payout % of earnings	5.69%
Terminal PE model	1	Tax Rate	18.23%
Choice of Terminal Value for		Fixed Debt/Assets Ratio	41.18%
Dividend Model: 1= Gordon		All Other Assets (Assume stays fixed \$ amount)	\$28,681.00
Growth, 2= Terminal PE model	2	Shares Outstanding	1000

GM's financial projections show stable but low growth, consistent with the mature automotive cycle. Sales are expected to rise slightly from \$189 billion in 2025 to \$195 billion in 2029, with EBIT maintaining a narrow range between \$21–22 billion. Earnings after tax grow modestly from \$10.9 billion to \$12.6 billion over the same period.

Dividends are forecast to increase gradually, from \$623 million to \$722 million, representing modest but consistent shareholder returns. GM's low reinvestment

rate and positive FCFE suggest steady cash generation but limited expansion potential.

	<u>2025</u>	<u>2026</u>	<u>2027</u>	<u>2028</u>	<u>2029</u>
	1	2	3	4	5
Sales	189097.03	190766.67	192451.05	194150.31	195864.57
COGS	146779.65	148075.65	149383.09	150702.08	152032.71
SGA	9504.18	9588.10	9672.76	9758.17	9844.33
Depreciation	11695.05	11798.31	11902.49	12007.58	12113.60
EBIT	21118.14	21304.60	21492.71	21682.48	21873.93
Interest	7733.59	7398.33	7060.12	6718.92	6374.71
Earnings before Tax	13384.55	13906.27	14432.59	14963.56	15499.22
Tax	2439.63	2534.72	2630.66	2727.44	2825.07
Earnings after Tax	10944.92	11371.55	11801.94	12236.13	12674.15
Dividends	623.27	647.56	672.07	696.80	721.74
Contribution to Retained earnings	10321.65	10723.98	11129.86	11539.33	11952.41

FCFE Valuation		Year Relative to Today	0	1	2	3	4	5
Determinants of FCFE								
EBIT*(1-T)				17268.90	17421.37	17575.19	17730.38	17886.93
Add Dep				11695.05	11798.31	11902.49	12007.58	12113.60
Less change in Gross Fixed Assets				1128.75	1138.72	1148.77	1158.92	1169.15
Less Change in NWC				108.43	109.38	110.35	111.32	112.31
FCFF				27726.77	27971.58	28218.56	28467.71	28719.07
Less After tax interest payments				6323.97	6049.83	5773.26	5494.25	5212.78
Add Net Borrowing				-3956.23	-3991.16	-4026.40	-4061.96	-4097.82
FCFE				17446.56	17930.59	18418.89	18911.51	19408.47
PV of Cash Flows				14,433.95	12,272.84	10,430.12	8,859.87	7,522.59

The FCFE model was chosen as the primary valuation framework because GM's dividend payments, while steady, represent only a small portion of its overall cash generation. The FCFE model captures both operating and financing effects—particularly relevant for GM, which frequently adjusts its capital structure through debt and share repurchases.

The DDM model, applied for comparison, yielded a lower intrinsic value of \$65.81 per share, also suggesting undervaluation but with less sensitivity to capital reinvestment assumptions. The difference between the two outcomes highlights how GM's reinvestment and financing policies enhance shareholder value beyond dividend payments alone.

Overall, both approaches support a Buy rating. GM's stock appears undervalued relative to its projected cash flows and steady operating performance. Despite slow top-line growth, GM's strong balance sheet, disciplined cost management, and continued shareholder returns justify a positive investment outlook. However, risks remain tied to macroeconomic cycles, EV transition costs, and supply chain volatility, which could pressure margins and valuation multiples.

Terminal Value (FCFE infinite growth model)	111,856.68
PV of Terminal Value (FCFE infinite growth model)	43,354.90
Terminal Value (Terminal P/E Model)	164,763.93
PV of Terminal Value (Terminal P/E Model)	63,861.40
PV Terminal Value	43,354.90
Sum of PVs (Total Intrinsic Value of Equity)	96,874.27
Shares outstanding	1,000.00
Value per share	\$ 96.87
Buy/Sell/Hold	Buy
Terminal Value (Dividend infinite growth model)	6,869.80
PV of Terminal Value (Dividend infinite growth model)	2,662.69
Terminal Value (Terminal P/E Model)	164,763.93
PV of Terminal Value (Terminal P/E Model)	63,861.40
PV Terminal Value	63,861.40
Sum of PVs (Total Intrinsic Value of Equity)	65,807.04
Shares outstanding	1,000.00
Value per share	\$ 65.81
Buy/Sell/Hold	Buy

Sensitivity Analysis

Another similar sensitivity analysis is being performed to GM. The areas include: (1) Cost of Equity & Long-term FCFE Growth Rate; (2) Cost of Equity & Long-term FCFE Growth Rate; and (3) Gross Fixed Asset(PPE)/Sales with FCFE value.

For GM, the theoretical value of the FCFE valuation is 96.87. Since GM is not classified as a high tech company, the P/E ratio is fairly low. Therefore, observe that there are some values that exceed the current market price and some are below, which is a positive indication of the overall valuation accuracy.

		DATA TABLE: SENSITIVITY OF FCFE STOCK VALUATION TO Cost of Equity AND LONG-TERM GROWTH RATE					
			Long-term FCFE growth ↓				
	96.87	1%	3%	4%	5%	6%	7%
Cost of Equity Capital	17%	55.88	65.13	70.82	77.46	85.31	94.72
	18%	50.40	58.25	63.02	68.52	74.94	82.52
	19%	45.64	52.36	56.39	61.00	66.32	72.52
	20%	41.46	47.26	50.70	54.60	59.06	64.20
	21%	37.79	42.82	45.78	49.11	52.88	57.19
	22%	34.54	38.93	41.49	44.35	47.57	51.23
	23%	31.65	35.50	37.74	40.21	42.99	46.10
	24%	29.07	32.47	34.43	36.59	38.99	41.67
	25%	26.76	29.78	31.50	33.39	35.48	37.81

DATA TABLE: SENSITIVITY OF Dividend STOCK VALUATION TO Cost of Equity AND LONG-TERM GROWTH RATE							
		Long-term Dividend growth ↓					
	65.81	7%	9%	14%	15%	16%	17%
Cost of Equity Capital	17%	3.67	4.73	14.37	23.45	61.82	-101.62
	18%	3.19	4.00	9.97	13.89	22.67	59.75
	19%	2.79	3.44	7.49	9.64	13.44	21.92
	20%	2.47	2.99	5.90	7.24	9.33	13.00
	21%	2.19	2.62	4.81	5.71	7.01	9.02
	22%	1.96	2.31	4.01	4.65	5.53	6.79
	23%	1.76	2.06	3.40	3.88	4.51	5.35
	24%	1.59	1.84	2.93	3.30	3.76	4.37
	25%	1.44	1.66	2.55	2.84	3.19	3.64

		FCFE Value
		96.87
	10%	56.02515773
Gross Fixed Assets (PPE)/Sales	15%	55.41256265
	20%	54.79996757
	25%	54.18737249
	30%	53.57477741
	35%	52.96218234
	40%	52.34958726
	45%	51.73699218
	50%	51.1243971
	55%	50.51180202

Note that the FCFE value decreases approximately proportionally with the PPE/Sales ratio, and similar traits were also shown on Tesla's table. This is because a higher PPE/Sales ratio implies that the firm must reinvest more heavily in fixed assets for every unit of sales it generates. In other words, as capital intensity rises, a greater portion of earnings is tied up in maintaining or expanding property, plant, and equipment rather than being available to equity holders as free cash flow.

Scenario Analysis

In the scenario analysis for General Motors, for the best-case scenario, sales growth is set at 7 percent with long-term FCFE growth of 8 percent and a cost of equity of 17 percent. It shows a situation where GM successfully leverages the industry shift toward electric vehicles and reduces perceived investor risk through stable cash flows and diversification. In the worst-case scenario, sales growth declines to 1 percent, long-term FCFE growth falls to 1 percent, and the cost of equity rises to 25 percent. The above inputs present a more adverse outlook in which the company faces weak industry growth and heightened risk premiums. The result is shown below:

Scenario Summary			
	Current Values: Worst Case		Best Case
Changing Cells:			
sales_growth	3.00%	1.00%	7.00%
LTgrowth_FCFE	3%	1%	8%
cost_equity	20.87%	25.00%	17.00%
Result Cells:			
Intrinsic_FCFE	96.87	\$ 26.76	\$106.23
Intrinsic_DDM	\$ 65.81	\$ 1.00	\$ 3.95

Sensitivity/Scenario Analysis Conclusion

First, the P/E ratio for GM is way lower compared to Tesla, which makes the sensitivity model more comparable to when it comes to valuation. Unlike Tesla, several of the simulated intrinsic values under the FCFE model exceed the current market price, in which it shows a closer alignment between valuation outcomes and market fundamentals. Note that this is consistent with GM's classification as a mature automaker with a lower P/E ratio where intrinsic value is more sensitive to changes in cost of equity and long-term growth assumptions. The results also show the impact of capital intensity where higher PPE/Sales ratios reduce free cash flow to equity by tying up earnings in fixed asset reinvestments. To summarize all the observations, GM's data provides a more realistic valuation framework. The outcomes have a positive contribution to the confidence of the valuation process.

Summary and Recommendation

Our valuation analysis compared Tesla Inc. (TSLA) and General Motors Company (GM) to determine the appropriate investment recommendation for each firm based on intrinsic value estimates derived from the Free Cash Flow to Equity (FCFE) model and the Fama–French three-factor model for cost of equity. Tesla's FCFE valuation produced an intrinsic value of \$12.94 per share, which is approximately

97% lower than its current market price of \$403.84, suggesting a severe overvaluation. Despite projecting an average sales growth rate of 6.5% and a cost of equity of 18.4%, all scenario outcomes—pessimistic, base, and optimistic—show Tesla's intrinsic value far below its market price. The company's P/E ratio of 233, declining market share from 70% to 39%, and limited free cash flow generation highlight the disconnect between its market valuation and underlying fundamentals. Tesla's high cost structure, intense industry competition, and capital-intensive operations further constrain profitability. While Tesla retains potential for growth in autonomous driving and energy storage, these opportunities remain speculative and highly uncertain. Therefore, Tesla's current market value reflects unrealistic growth assumptions driven by investor sentiment rather than operational performance, leading us to issue a sell recommendation.

Conversely, General Motors (GM) demonstrates solid fundamentals and substantial undervaluation. Using the FCFE model, GM's intrinsic value was estimated at \$96.87 per share, compared to a market price of \$53.27, indicating an 82% upside potential. The supporting DDM model produced a similar valuation of \$65.81, confirming consistency across methodologies. GM's cost of equity of 20.87% and conservative sales growth projection of 0.88% were derived from the Fama–French model, yet even under these assumptions, the stock remains undervalued. GM's P/E ratio of 4.65 is well below the industry average, reflecting significant market discount, while its efficient cost control (COGS at 77.6%, SG&A at 5.0%) and moderate leverage (debt-to-asset ratio of 41.2%) support financial stability. Scenario analysis results show strong resilience, with intrinsic values ranging from \$26.76 to \$106.23, providing an attractive risk–reward balance. GM's strategic investments in electric and autonomous vehicles, coupled with a 5.7% dividend yield, position it as a reliable value-oriented investment. Overall, while Tesla appears grossly overvalued with high downside risk, GM's valuation is well supported by fundamentals and conservative growth expectations. Accordingly, we recommend selling Tesla and buying General Motors, as GM offers sustainable returns and a realistic long-term growth outlook consistent with market fundamentals.

References

1. <https://ycharts.com/>
2. <https://finance.yahoo.com/>
3. <https://www.cnbc.com/2025/01/28/general-motors-gm-earnings-q4-2024.html>
4. <https://www.prnewswire.com/news-releases/honda-motor-co-ltd-hmcnyse-announced-its-consolidated-financial-results-for-the-fiscal-first-quarter-ended-june-30-2025-302523182.html>
5. <https://www.investopedia.com/articles/markets/021416/how-toyota-makes-money-tm.asp>
6. <https://www.investopedia.com/terms/p/porter.asp>
7. <https://money.usnews.com/investing/articles/upstart-tesla-competitors-to-watch>
8. <https://investingnews.com/daily/resource-investing/base-metals-investing/nickel-investing/top-electric-vehicle-stocks/>